

Variation, independence and multiple testing

What would this result look like if nothing biological had changed?

By the end, you should be able to...

Identify the experimental unit and a confounded comparison.

Interpret a p-value without calling it the probability the hypothesis is true.

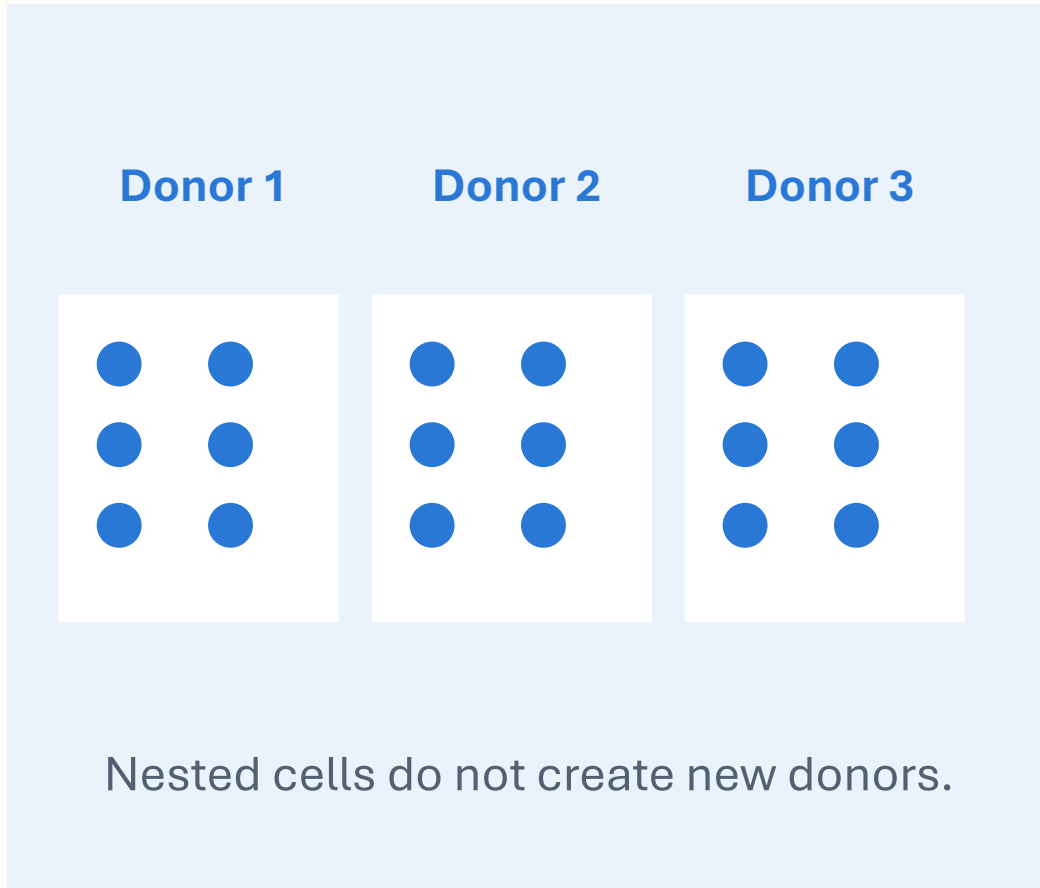
Apply a small BH correction and explain its limits.

Reconnect with the biology

Technical repeats share biological origins; more observations are not automatically more independent evidence.

Think silently -> explain to a neighbour -> check the measurement.

Variation has different sources

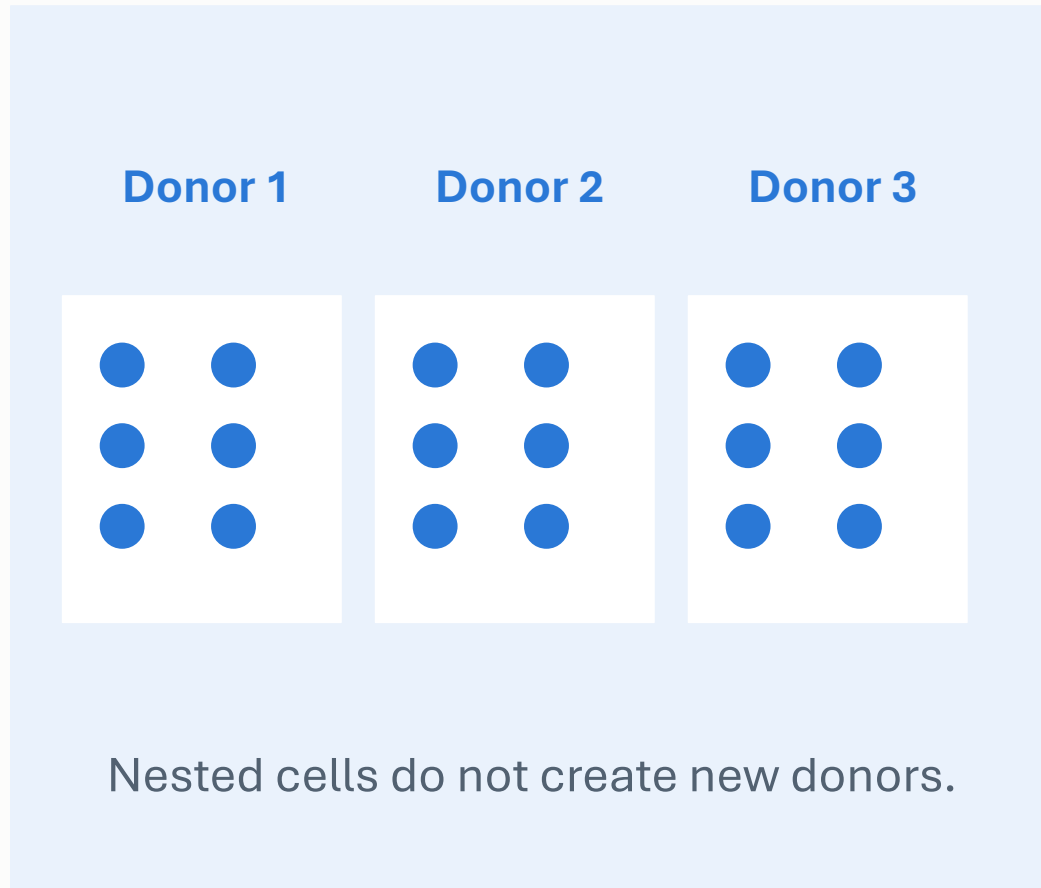


Biological variation: independently sampled experimental units can differ.

Technical variation: measurement or processing can vary.

The design determines which differences can be attributed to treatment.

Count independent units, not rows



Three donors, each with 100 measured cells, still give three donors.

For donor-level claims, uncertainty must account for donor-to-donor variation.

Cells can be valid observations for other questions; name the question first.

Confounding is a design problem

Bad design	Better design
Batch A: treatment only	Batch A: treatment + control
Batch B: control only	Batch B: treatment + control

All treated samples in batch A and all controls in batch B confound treatment and batch.

A statistical adjustment cannot create the missing comparison.

Balance conditions across batches before data collection where feasible.

What a p-value actually asks

01

Assume the stated null model and its assumptions.

>

02

How probable are data at least as extreme by the chosen test statistic?

>

03

It is not $P(\text{the null is true})$ and not a measure of effect size.

Effect, uncertainty and p answer different questions

Size and uncertainty

Compatibility with a null

Effect size: how large is the estimated difference?

Uncertainty interval: which values are compatible with the model and data?

p-value: how incompatible is the chosen statistic with the null model?

Testing many nulls generates false positives

$$1,000 \times 0.05 = 50$$

Expected, not guaranteed

For 1,000 valid null tests at alpha 0.05, expected false positives are 50.

This expectation does not require all tests to be independent.

The observed count varies; no single run must equal 50.

BH controls a family-level error quantity

Rank	p	Threshold, $q=0.05$
1	0.001	0.0125
2	0.010	0.0250
3	0.040	0.0375
4	0.200	0.0500

Sort p-values; compare $p(i)$ with $i \times q / m$.

Select through the largest passing rank in the step-up rule.

FDR is an expected false-discovery proportion under assumptions, not each gene's error probability.

A negative control tests the pipeline

01

Simulate a known null or use an appropriate design-respecting permutation.

>

02

Repeat the experiment rather than pick one convenient random seed.

>

03

A control failing to reject does not prove every assumption is correct.

Worked example: a four-test BH family

$p = 0.001, 0.010, 0.040, 0.200$; target FDR $q = 0.05$.

Predict the result and one limitation before calculating.

Work through the reasoning

BH thresholds are 0.0125, 0.0250, 0.0375, 0.0500.

The largest passing rank is 2, so reject tests 1 and 2.

Monotone adjusted p-values are 0.004, 0.020, about 0.0533, 0.200.

Pause: make a defensible claim

An analyst calls 100 cells per donor 100 independent replicates. There are two donors per condition. What is the donor-level sample size per condition?

First answer alone. Then compare the evidence, not just the conclusion.

A defensible response

Two donors per condition. Treating all cells as independent for that comparison is pseudoreplication and can underestimate uncertainty.

Today's practical

W06: simulate uniformly distributed valid null p-values, compare raw and BH discoveries across repeated runs, and identify an intentionally wrong independence claim.

Minutes	Action
0-10	Predict + set up
10-30	Run + inspect + check
30-40	Explain + compare
40-45	Save + submit / exit

Five terms to use precisely

- 1 null
- 2 p-value
- 3 effect size
- 4 confounding
- 5 FDR

Exit ticket

What did we actually measure or calculate?

What is one assumption or limitation of that result?

What check would you perform before sharing it?

Go further only after the core is clear

Optional: show why dependence affects a multiple-testing guarantee; do not memorize a proof.

Not required for the core grade. No advanced model training is expected.

Read the evidence, not only the slide

S07 American Statistical Association. Statement on statistical significance and p-values (2016).

S08 Benjamini and Hochberg (1995). Controlling the false discovery rate: a practical and powerful approach to multiple testing.

S10 DESeq2: Analyzing RNA-seq data with DESeq2, release vignette.

S12 Squair et al. (2021). Confronting false discoveries in single-cell differential expression.